Assignment

Date

Q1. A survey of a class of 30 KU graduates showed that 4 graduated with First Class Honors 10 graduated with Second Class Honors and 16 graduated with Third Class Honors. What is the probability that a graduate selected from this group graduated with First Class Honors.

Solution :-

$n(S) = 30$ (Total graduates)

$n(A) = 4$ (First Honor graduate)

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{30} = \frac{2}{15}$$

$$P(A) = \frac{2}{15}$$

The probability is $\frac{2}{15}$

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Q2. In a sample of 50 boys (people), 21 had type O blood 22 had type A blood, 5 had type B blood and 2 had type AB blood. Set up a frequency distribution and find the following probabilities.

a. A person has type O blood.
b. A person has type O or type blood.
c. A person has neither type A nor type O blood
d. A person does not have type AB blood.

Solution

Frequency Distribution

Blood type	Frequency
O	21
A	22
B	5
AB	2
Total	50

(b) A person has type A or type B blood.

$$P(A \text{ or } B) = \frac{22+5}{50} = \frac{27}{50}$$

$$P(A \text{ or } B) = \frac{27}{50}$$

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c. A person has type O blood

$$P(O) = \frac{21}{50}$$

c. A person has neither A nor O type blood

$$n(\text{neither A nor O}) = 50 - 21 - 22 = 50 - 43 = 7$$

$$P(\text{neither A nor O}) = \frac{7}{50}$$

d. A person does not have type AB blood.

$$n(\text{not AB}) = 50 - 2 = 48$$

$$P(\text{not AB}) = \frac{48}{50} = \frac{24}{25}$$

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Q3. In a high school graduating class of 100 student, 54 studied mathematics, 69 studied history and 35 studied both mathematics and History. If one of these students is selected at random, find the probability that

a. The student took mathematics or History.
b. The student did not take either of these subject
c. The student took history but not maths.

Solution :-

a. The student took mathematics or History

$$n(M \cup H) = n(M) + n(H) - n(M \cap H) = 54+69-35 = 88$$

$$P(M \cup H) = \frac{88}{100} = \frac{22}{25}$$

b. The student did not take either subject

$$n(\text{neither}) = 100 - 88 = 12$$
$$P(\text{neither}) = 12/100 = 3/25$$

c. The student took History but not Maths.

$$n(H \setminus M) = n(H) - n(M \cap H) = 69 - 35 = 34$$

$$P(H \setminus M) = 34/100 = 17/50$$

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Q4. Suppose it is on records that out 25 fresh students who offered STAT 311 in the 2022/2023 academic session at the university, 10 are from Statistics department, 8 from electrical engineering department while the remaining 7 were from physics department respectively. What are the chances of having STAT 311 students in the 2023/2024 academic session from:

a. Statistics department
b. Physics department
c. Electrical Engineering department
d. Hence or otherwise, what approach of probability is this?

Solution:-

a.) From Statistics Dept:-

$P(S) = 10/25 = 2/5$

b.) From Physics Dept:-

$P(P) = 7/25$

c.) From Electrical Engineering Dept

$P(E) = 8/25$

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d. What approach of probability is this?

This is empirical (experimental) Probability (also called Relative frequency Approach), because the probabilities are based on recorded data from past observations (2022/2023) not on theoretical assumptions.

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Q5. A man owns a house in town and a cottage in the country. In any one year the probability of the house being burgled is 0.01 and the probability of the cottage being being burgled is 0.05. In any one year what is the probability that:

a.) both will be burgled

b.) One or the other (but not both) will be burgl

Solution :-

$P(H) = 0.01$

$P(C) = 0.05$

a. Both will be burgled

$P(H \cap C) = P(H) \times P(C) = 0.01 \times 0.05$

$P(H \cap C) = 0.0005$

b. One or the other (not both)

$P(\text{exactly one}) = P(H \cap C') + P(H' \cap C)$

$P(\text{exactly one}) = P(H)(1-P(C)) + (1-P(H)) \cdot P(C)$

$P(\text{exactly one}) = 0.01(1-0.05) + (1-0.01)(0.05)$

$P(\text{exactly one}) = 0.0095 + 0.0495$

$P(\text{exactly one}) = 0.059$

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Q6. World wide insurance company found that 53% of the residents of a city had homeowner's insurance (H) with the company. Of these clients, 27% also had automobile insurance (A) with the company. If a resident is selected at random, find the probability that the resident had both homeowner's and automobile insurance with world wide insurance company.

Solution:-

$P(H) = 0.53$

$P(H \text{ and } A) = 0.27$ (Those who have automobile insurance will have homeowner's insurance → given)

For those who have both (selected at random)

$P(H \cap A) = P(H) \cdot P(H \text{ and } A)$

$P(H \cap A) = 0.53 \times 0.27$

$P(H \cap A) = 0.1431$

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Q7. At a gathering consisting of 10 men and 20 women, two door prizes are awarded. The winning ticket is not replaced. Find the probability that

a. both prizes are won by men
b. both prizes are won by women
c. a men wins one and a women wins one
d. would you consider this event in a, b, and c likely or unlikely to occur?

Solution :-

Total people = 10 men + 20 women = 30

a Both prizes are won by men.

First prize winner (men) = $\frac{10}{30} = \frac{1}{3}$

Now one prize distributed so one man is deducted

Second prize winner (men) = 9/29

both prizes winner (men) = $1/3 \times 9/29 = 3/29$

both prizes won by men = 0.103

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b. Both prizes won by women :-

First prize winner (women) = 20/30 = 2/3

Second prize winner (women) = 19/29 = 19/29

Both prizes winner (women) = 2/3 × 19/29 = 38/87

Both prizes winner (women) = 0.437

c) A man wins and a women wins

1.) Man first, then women

$$P(MW) = 10/30 \times 20/29 = 200/870$$

2.) Women first, then men

$$P(WM) = 20/30 \times 10/29 = 200/870$$

$$P(\text{one man, one women}) = \frac{200}{870} + \frac{200}{870} = \frac{400}{870}$$

$$P(\text{one man, one women}) = 0.460$$

d) Are these events likely on unlikely.

- Both men : 0.103 ⟶ unlikely
- Both women : 0.437 ⟶ fairly likely
- One man and one women : 0.46 ⟶ most likely

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Q8 A bag contains 7 white, 3 Red, and 5 Black balls. Three are drawn without replacement. Find probability that:

a. No ball is red
b.) exactly one is Red
c. At least one is red
d.) All are of the same colour
e. No two are of the same colour.

Solution :-

Total balls = 7+3+5 = 15

Number of ways to choose 3 balls = $^{15}C_3 = 455$

Sample space is = 455.

a. No ball is red

No ball is red means we will select 3 balls from 12 balls (deducted all red balls)

No. of ways to choose 3 balls now = $^{12}C_3 = 220$

P(no red) = 220/455 = 0.4835

b. Exactly one is red

choose 1 ball from 3 (red) and remaining two from 12 ba

No. of ways to choose exactly 1 red = $^{3}C_1 \cdot {}^{12}C_2 = 198$

P(exactly one red) = 198/455 = 0.435

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c. At least one is red
Subtract the probability of getting no red balls from 1

$= 1 - P(\text{no red}) = 1 - 0.4835$

$P(\text{exactly one red}) = 0.5175$

d. All are of the same colour :-

No of ways to select all red $= {}^3C_3 = 1$

No of ways to select all white $= {}^7C_3 = 35$

No of ways to select all black $= {}^5C_3 = 10$

$P(\text{all of same colour}) = \frac{1+35+10}{455} = 0.1010$

e. No two are of the same colour :-
We have to select one white, one red and one black

${}^7C_1 \cdot {}^3C_1 \cdot {}^5C_1 = 105$

$P(\text{one of each colour}) = \frac{105}{455} = 0.2308$

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Q9. A KU student is enrolled in a course in Geology and an course in Statistics. The probabilities that the student will pass Geology, Statistics or both subjects are, respectively, $P(G) = 0.8$, $P(S) = 0.7$ and $P(G \cap S) = 0.56$

a. What is probability that the student will pass Geology given that the student passes Statistics?

b. Are the events G and S independent?

Solution:-

a. Probability that student passes geology already passed Statistics.

Means we will used all statistics passed prob as sample space and pass in both as seleted event

$$P(G|S) = \frac{P(G \cap S)}{P(S)} = \frac{0.56}{0.7} = 0.8$$

$$P(G|S) = 0.8$$

b. Are G and S independent?

Two events are independent if

$$P(G \cap S) = P(G) \cdot P(S)$$

$$P(G \cap S) = 0.8 \times 0.7$$

$$P(G \cap S) = 0.56$$

Yes both events are ~~or~~ independent

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Q10. A maintenance firm has gathered the following information regarding failure mechanisms for air conditioning systems:

Evidence of electrical failure	Evidence of gas failure		
	Yes	No	Total
Yes	55	17	72
No	32	3	35
Total	87	20	

The units without evidence of gas leaks or electrical failure showed other types of failures. If this is representative sample of AC failure, find the probability.

a. That failure involves gas leak
b. That there is evidence of electrical failure given that there was a gas leak.
c. That there is evidence of a gas leak given that there is evidence of electrical failure.

Solution :-

Sample size is = $(55+17) + (32+3) = 107$

a) Probability the failure involves a gas leak.

$$P(\text{Gas}) = 87/107 = 0.8131$$

b) Probability that if there is electrical failure then there is gas leak also

$$P(\text{Electrical} \mid \text{Gas}) = \frac{P(\text{Electrical} \cap \text{Gas})}{P(\text{Gas})} = \frac{55}{87}$$

$$P(\text{Electrical} \mid \text{Gas}) = 0.6322$$

c. Probability that if there is gas leak then there is electrical failure also

$$P(\text{gas} \mid \text{electrical}) = \frac{P(\text{gas} \cap \text{Electrical})}{P(\text{electrical})}$$

$$P(\text{gas} \mid \text{electrical}) = \frac{55}{72}$$

$$P(\text{gas} \mid \text{electrical}) = 0.7638$$

Q11. A building has 60 occupants consisting of 15 women & 45 men. The men have probability 1/2 of being colorblind and the women have probability 1/3 of being colorblind.

a.) Suppose you choose uniformly at random a person from the 60 in building. What is the probability that the person will be colorblind?

b.) Determine the conditional probability that you chose a woman given that the ~~opp~~ occupant you chose is colorblind.

Solution :-

Total occupant = 60

Women = 15

Men = 45

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a. Probability that a randomly chosen person is colorblind.

$$P(\text{colorblind}) = P(\text{men})P(\text{colorblind} \mid \text{men}) + P(\text{women})\,P(\text{colorblind} \mid \text{women})$$

$$= \frac{45}{60} \times \frac{1}{2} + \frac{15}{60} \times \frac{1}{3}$$

$$= \frac{3}{8} + \frac{1}{12}$$

$$P(\text{colorblind}) = 0.458 \quad \text{or} \quad 11/24$$

b. Conditional probability that the chosen person is a women given colorblind

$$P(\text{women} \mid \text{colorblind}) = \frac{P(\text{women and colorblind})}{P(\text{colorblind})}$$

$$P(\text{women} \mid \text{colorblind}) = \frac{P(\text{women}) \cdot P(\text{colorblind} \mid \text{women})}{P(\text{colorblind})}$$

$$P(\text{women} \mid \text{colorblind}) = \frac{15/60 \times 1/3}{11/24} = \frac{1/12}{11/24}$$

$$P(\text{women} \mid \text{colorblind}) = \frac{2}{11}$$

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Q12 The probability is 1 % that an electrical connector that is kept dry fails during the warranty period of a portable connector. If the connector is ever wet, the probability of a failure during the warranty period is 5 %. If 90 % of the connectors are kept dry and 10 % are wet, what proportion of connectors fail during the warranty period?

Solution :-

$P(\text{fail} \mid \text{dry}) = 0.01$

$P(\text{fail} \mid \text{wet}) = 0.05$

$P(\text{dry}) = 0.9$

$P(\text{wet}) = 0.1$

$P(\text{fail}) = P(\text{dry})\,P(\text{fail} \mid \text{dry}) + P(\text{wet})\,P(\text{fail} \mid \text{wet})$

$P(\text{fail}) = 0.9 \times 0.01 + 0.1 \times 0.05$

$P(\text{fail}) = 0.014$

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for smokers probability of dying is 0.05

Q8. Ten percent of a company's life insurance policyholders are smokers. The rest are non smokers, the probability of dying during the year is 0.01. Given that a policy holder has died, what is the probability that the policy holder was a smoker.

Solution :-

$P(\text{smoker}) = 0.1$

$P(\text{non-smoker}) = 0.9$

$P(\text{die} \mid \text{smoker}) = 0.05$

$P(\text{die} \mid \text{nonsmoker}) = 0.01$

Now,

$$P(\text{smoker} \mid \text{died}) = \frac{P(\text{smoker})\,(P(\text{died} \mid \text{smoker}))}{P(\text{died})}$$

$$= \frac{P(\text{smoker})\,P(\text{died} \mid \text{smoker})}{P(\text{smoker})\,P(\text{died} \mid \text{Smoker}) + P(\text{nonsmoker})P(\text{died} \mid \text{nonsmoker})}$$

$$= \frac{0.1 \times 0.05}{0.1 \times 0.05 + 0.9 \times 0.01}$$

$$= 0.357$$

$P(\text{smoker} \mid \text{died}) = 35.7\%$

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Q14. Samples of laboratory glass are in small, light packaging or heavy, large packaging. Suppose that 2 and 1 % of the samples shipped in small and large packages, respectively, break during transit. If 60% of the samples are shipped in large packages and 40% are shipped in small packages, what proportion of samples break during shipment?

Solution:-

$P(\text{break} \mid \text{small}) = 0.02$

$P(\text{break} \mid \text{large}) = 0.01$

$P(\text{small}) = 0.4$

$P(\text{large}) = 0.6$

$$P(\text{break}) = P(\text{small})P(\text{break} \mid \text{small}) + P(\text{large})P(\text{break} \mid \text{large})$$

$$= 0.4 \times 0.02 + 0.6 \times 0.01$$

$$= 0.014$$

$$P(\text{break}) = 1.4\ \%$$

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Q 15. A test for a certain rare disease is assumed to be correct 95% of the time: if a person has the disease, the test results are positive with probability 0.95, and if a person does not have the disease the test results are negative with probability 0.95. A random person drawn from a certain population has probability 0.001 of having the disease. Given that the person just tested positive. what is the probability of having the disease.

Solution :-

Let;

D = person has disease

$\bar{D}$ = person doesn't has disease

$+$ = test is positive

$P(D)$ $P(\text{Disease}) = P(\text{person with disease}) = 0.001$

$P(\bar{D})$ $P(\text{person with no disease}) = 0.999$

$P(+|D)$ $P(\text{test is positive}) = 0.95$

$P(-|\bar{D})$ $P(\text{test is not positive}) = 0.05$

$$P(D|+) = \frac{P(D)P(+|D)}{P(D)P(+|D) + P(\bar{D}) \cdot P(+|\bar{D})}$$

$$= \frac{(0.001)(0.95)}{(0.001)(0.95) + (0.999)(0.05)}$$

$$P(D|+) = 0.0186$$

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